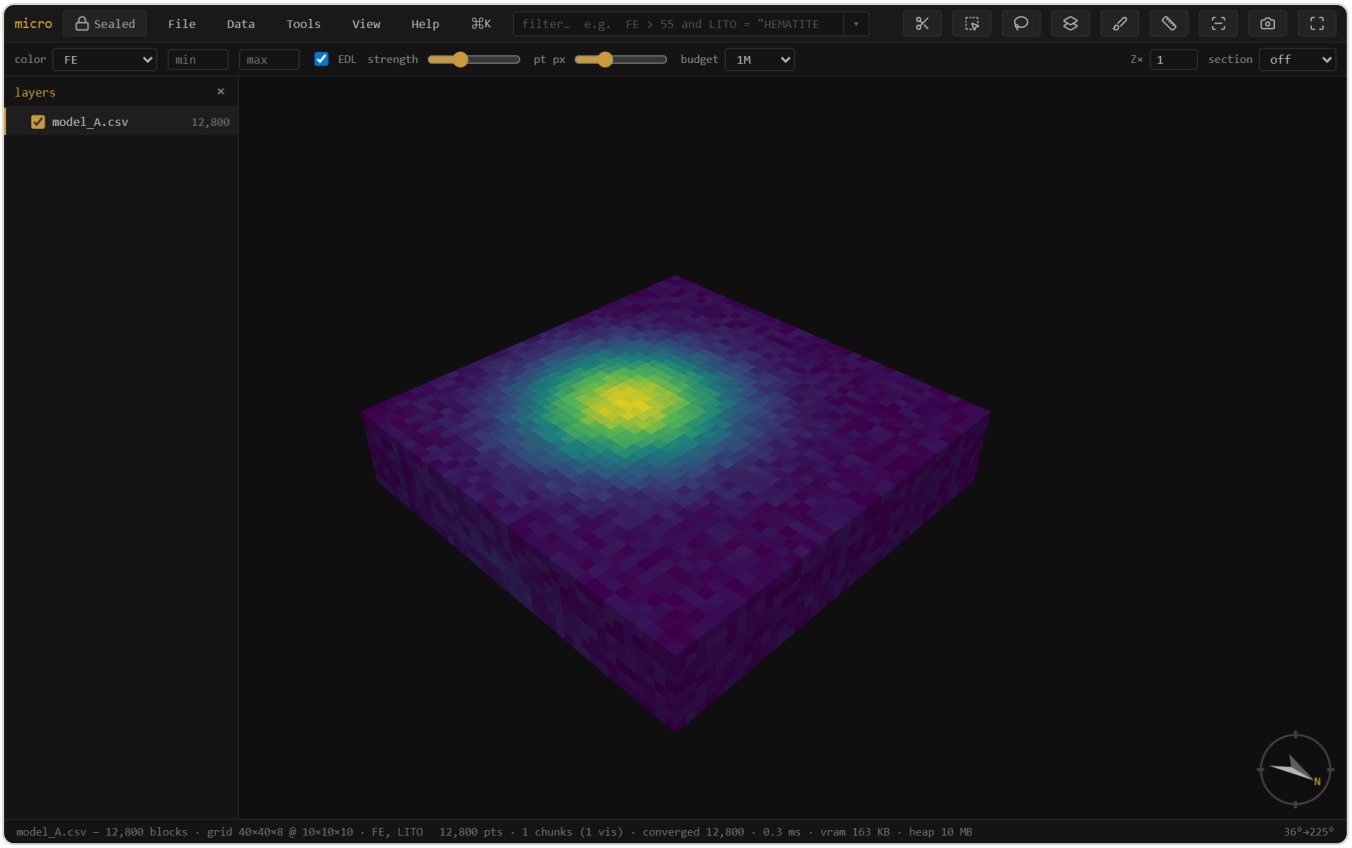
micro

A streaming, offline workbench for massive point clouds and block models.

🔒 **Sealed** — no network, nothing leaves your machine · single self-contained HTML file · download micro.html · open the app ↗

User documentation · Auditable / Geoscientific Chaos Union · gentropic.org/micro

micro opens a point cloud, block model, grid, or set of drillholes — however large — in about a second, with no import step, and lets you view, query, derive, join, reconcile, validate, and produce report-ready figures from it. Every derived column and layer carries a provenance trail, and nothing ever touches the network.



A 12,800-block iron model coloured by grade — the high-grade shoot in yellow. Loaded, inferred, and rendered from a plain CSV with no import step.

Contents

1 What micro is

micro is a single self-contained HTML file — a **viewer that grew into a workbench**. It streams massive spatial-element files (never holding them fully in memory), so the first look is near-instant at any size and the picture densifies while you orbit.

It reads:

- **Point clouds** — LAS, PLY, XYZ / PTS / DAT dumps
- **Block models** — CSV / TXT, Datamine `.dm`, and **Parquet**, sub-blocked included
- **Grids** — GeoTIFF, Surfer `.grd`, ESRI ASCII `.asc`
- **Meshes** — `.msh` (Leapfrog), OBJ, PLY-with-faces, LFM, Datamine wireframe pairs
- **Drillholes** — collar + survey + interval tables (desurveyed on load); a set can carry **several interval tables** (assays, lithology, geotech) sharing one geometry — see Drillholes

Beyond viewing, it does the everyday resource-geology work: filter and section; derive columns by expression, lookup, broadcast, resampling, or interpolation; domain by a solid; join and reconcile model versions; grade–tonnage, swath, and validation analysis; and figure export — all against the source file, all with a provenance trail, all offline.

> **Tip —** The fastest way to reach anything is the **command palette**: `Ctrl/⌘K` (or the `⌘K` button). It searches every action and shows only what applies to the layer you have selected.

2 Getting started

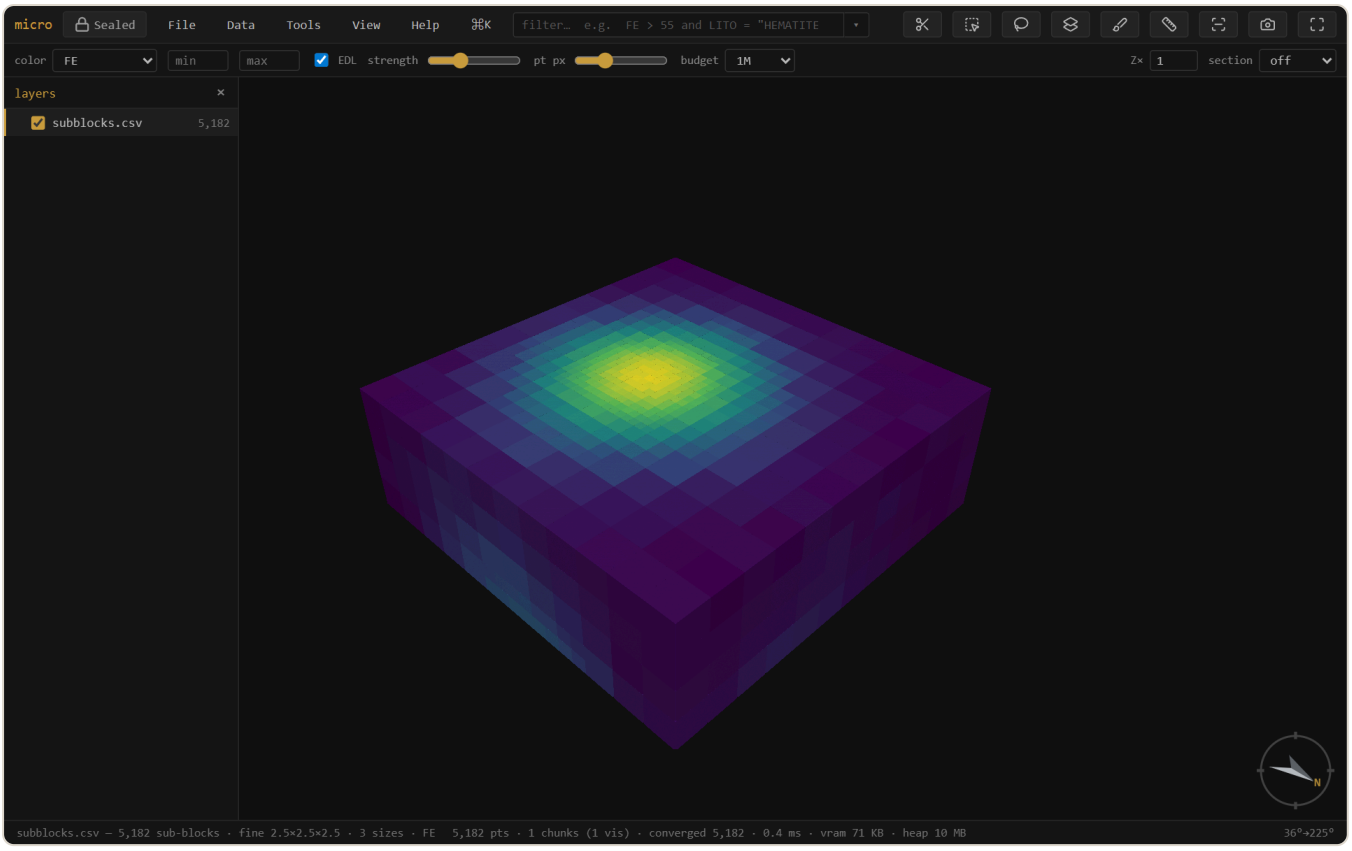
Opening data

Drag a file (or several) anywhere onto the window, or use **File** → **Open…**. Several files at once become separate **layers**. The open dialog shows each file's detected role — an **Open as** ▾ on every row lets you override it (a block model as points, a PLY mesh as points, a delimited file as collar / survey / intervals), and drillhole trios assemble themselves from those roles.

No data handy? The empty state offers ▸ **open the demo project** — one click generates a full scene locally: a 1.8M-block iron model with realistic bimodal grades, a drillhole set threading the ore, pre-mining **topography**, three **pit-phase shells**, and the **ore-shell** mesh the holes chase — sectioned open on the lens, with a grade–tonnage curve already computed. Every workflow in this manual can be tried on it, and **File** → **Save project as…** writes it to a real folder you can inspect: plain CSVs, JSON manifests, no black box.

Sub-blocked models

micro reads **sub-blocked** (octree) block models — where cells are refined into smaller boxes near a boundary or a high-grade zone — and renders **each block at its true size**, straight from the file. CSV/Parquet models with per-block `dX/dY/dZ` columns and Datamine `.dm` models with per-record `XINC/YINC/ZINC` are detected automatically; the model streams, picks, filters, and colours like any other. Join and reconcile handle them by **volume** — see Join & reconcile.



A sub-blocked (octree) model — coarse parent blocks away from the shoot, refined into progressively smaller boxes through the high-grade core, each drawn at its own size.

> **Note —** Detection assumes an **axis-aligned** model with power-of-two (octree) refinement — the standard export. A **rotated** model currently opens as its centroids (points); interpreting a rotated model as boxes (enter or measure its azimuth) is on the roadmap.

The layers panel

`p` toggles the layers panel. Each opened dataset is a layer with its own colour, filter, and visibility. Layers can be grouped and reordered by dragging; `[` `]` step the active layer, `h` hides/shows it, `⇧H` **solos** it (press again to restore what was visible). Drillhole sets appear as their own group with collar and survey rows — see Drillholes.

Right-click a layer for its menu, organized top to bottom: **identity** (zoom, rename, reinterpret) · **inspect & analyze** (attribute table, filter, stats, grade–tonnage, swath, file info, lineage, export rows) · **derive & store** (interpolate, materialize, store as Parquet, index) · **display** (opacity, edges, sections, hide) · **arrange** (move, copy into project, remove) · **Properties…**.

`F2` renames the active layer. **Rename layers…** (on a group, a multi-selection, or the palette) batch-renames: one-click **strip the common prefix/suffix**, or find→replace with optional **regex**, with a live before→after table.

Getting around

Drag to orbit, right-drag (or shift-drag) to pan, wheel to zoom. `f` fits the data. `n s e w` look level that way, `d` is plan, `o` toggles orthographic (needed for a true scale bar). Click a block or point to **pick** it — the record panel shows its full source row, calculated and derived columns included, with a **fields** picker to trim it to the columns you care about.

Vertical exaggeration — the **Z×** box on the instrument rail (or View → Vertical exaggeration) stretches the whole scene vertically so subtle relief or a flat-lying ore lens reads. It's a property of the *view*, not one layer: every layer stretches together and stays registered, and it's display-only — picking, measuring, and section cutting all stay in true coordinates. It travels with bookmarks and scenes.

3 Viewing & sections

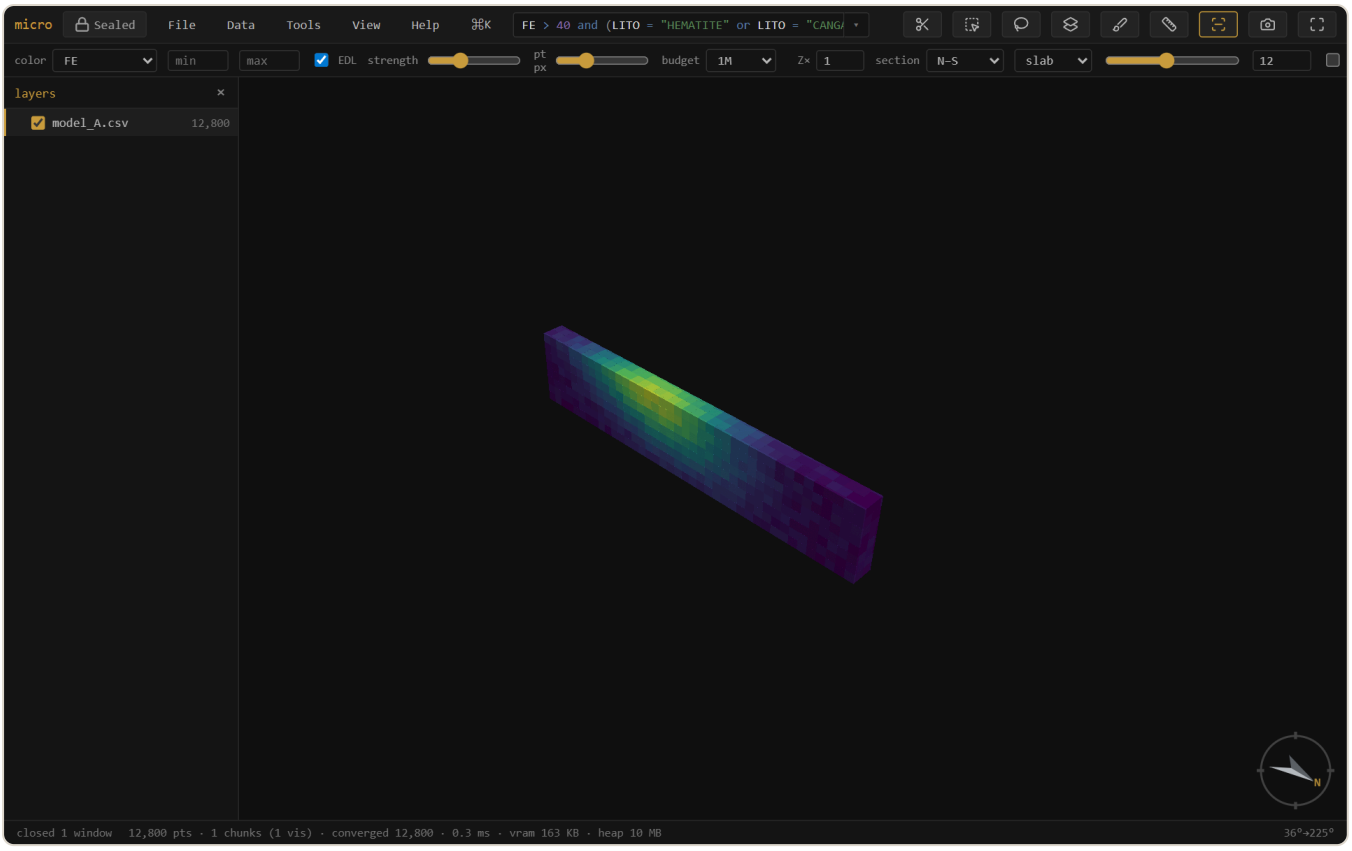
Colour

The **colour** dropdown drives the active layer: by elevation, a numeric **channel** (a grade), a **category** (a rock type), a painted column (🖌), a materialized column (%), or a calculated column (ƒ — see the ƒ workbench). Pick a ramp per layer (viridis · spectral · magma · turbo · greys) in **Properties** → **style**; the **min/max** boxes clip the scale so outliers stop eating the ramp. On a grade channel, double-click the histogram to add **breakpoints** — **classed** turns them into a solid-colour grade-shell legend.

Sections & measure

A section is a live cut, and on block models it is a **true** cut: each block is clipped analytically against the slab faces, so the cut paints a continuous wall — with the block's colour — instead of a ragged centroid cull, and the wall itself is pickable.

Meshes and surfaces draw their trace on the wall. A sectioned mesh doesn't hide the cut face: it renders as its **intersection trace** — a closed ore shell becomes an outline ring around the grade zone it encloses, a topo surface a thin line across the top — flattened onto the section face like a drafted section plot. The trace width follows the section-width control. A layer that should stay whole (topo for context, the drillholes threading a cut model) sets right-click → **Cut by sections** → **don't cut**.



A thin N–S slab. The cut face is a continuous coloured wall — blocks are clipped against the plane, not dropped by centroid — and clicking the wall picks the block behind it.

Pick a preset (plan · N–S · E–W), press `k` and drag a line (⇧ snaps 15°, ⇧⌥ 5°), or type a plane's dip-direction + dip. `x` toggles it, `l` faces it, `⇧L` faces its back. `,` `.` step one slab at a time, and **Ctrl+wheel** scrubs the position continuously; **Alt+wheel** thickens or thins the slab. **Follow** makes the camera ride along as you scrub; **2D** locks the view flat onto the section — drags pan instead of orbit — for section-by-section interpretation. `m` then two clicks **measures** 3D · plan · Δz between records. Per layer, **Cut by sections** picks how the section treats it: **follow** (the slab), **keep front only** / **keep behind only** (a half-space split at the plane), or **don't cut**.

Block edges

`⇧B` (or View → block edges) outlines every block — the classic wireframe-over-solid look for checking sub-blocking or lattice alignment. Each layer can override the global toggle from its context menu (always / never / follow).

The attribute table & the scan

`t` opens the layer's rows as a draggable, layer-pinned window — windowed reads, so a 1.8M-row model scrolls cold. **matches** shows only the filtered rows; click a row to pick it in the scene. In **Properties → columns**, **scan columns** streams the whole table once to fill n · mean · std · quantiles · histogram per column — the scan sees calculated and derived columns too.

The stats table

Stats… (right-click a layer, or the palette) is the grouped summary report: pick columns, an optional **group by** (the category channel or any painted column — *assign domains, then stats by domain* is the classic resource table), and an optional **weight expression** (`SG` for density-weighted means — quantiles stay unweighted and the window says so). Same chassis as grade–tonnage and swath: scope (all / filt / sel), Run, ⧉ **table** TSV copy, **save as recipe…**, and the setup persists with the project.

4 The filter & its widgets

`/` focuses the filter box. It speaks a familiar expression language over **every** column — source, calculated, painted, materialized, joined:

```
FE > 55 and LITO = "HEMATITE"
AU_OK between 0.5 and 3  or  DOMAIN in ("ORE", "HALO")
SIO2 > 10 and not (LITO = "CANGA")   # comments are fine
```

Matching elements isolate; everything else drops out of the render, the table, and any analysis. `Enter` applies, `Esc` clears. The box is **syntax-highlighted** as you type, autocomplete offers columns, functions, and category values, and unknown names get a *did-you-mean*. The full language — operators, functions, quoting, comments — is in the reference.

On big files the filter is engineered to be **live**: Parquet models skip whole row-groups their footers prove can't match, CSV and `.dm` files skip stretches via the project's band index, and after one complete scan the touched columns stay **pinned in memory** (see the project store) — so dragging a widget re-filters millions of blocks in milliseconds.

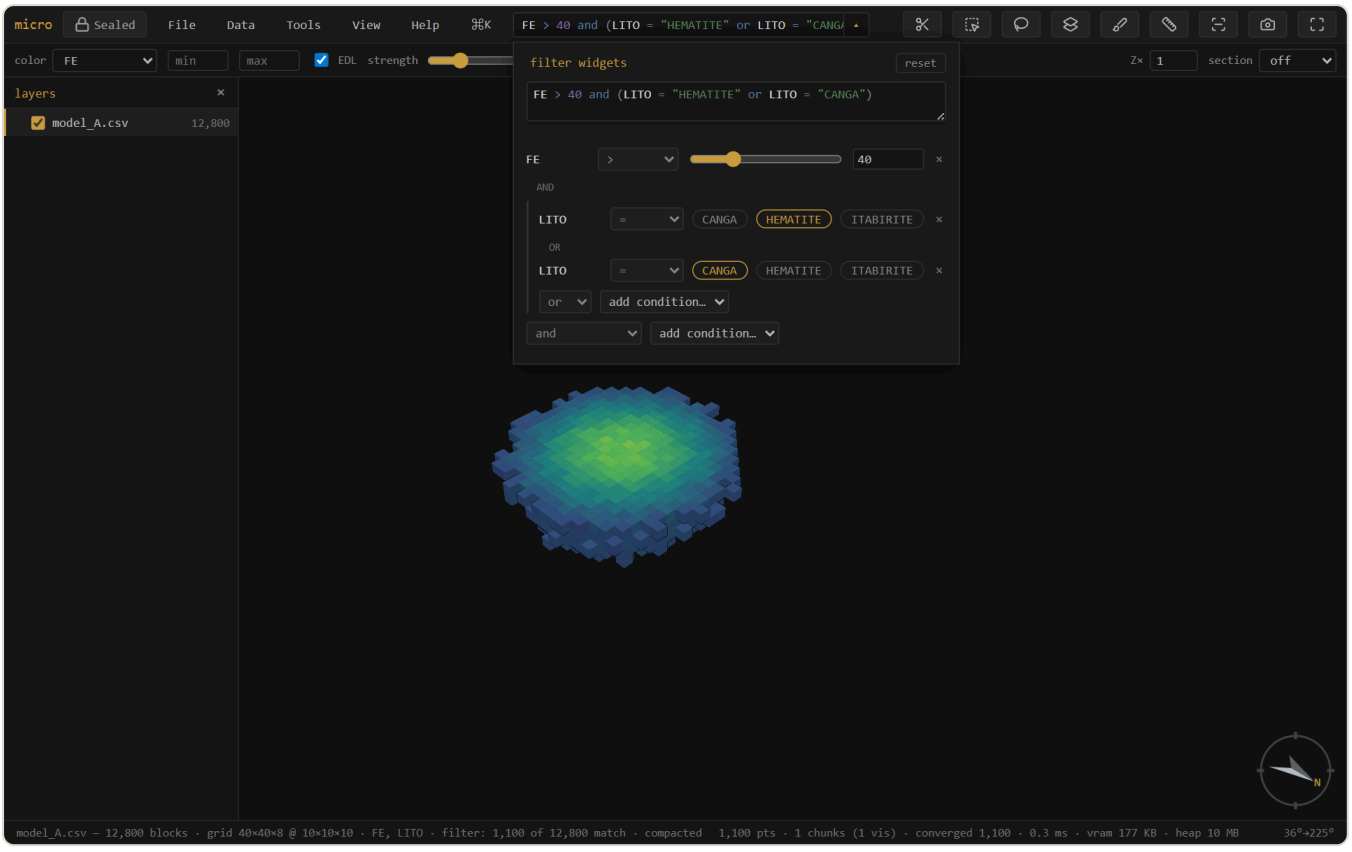
The filter drawer — the expression as live controls

The ▾ button glued to the filter box opens the **drawer**: your current expression projected as widgets. The expression stays the single source of truth — every control rewrites its own span of the text, so the box updates as you drag and your formatting survives.

- **Numeric comparisons** become sliders with a number box; `between` becomes a two-slider range row. The **operator is a dropdown** (`> ≥ < ≤ = ≠ between`) — switching to `between` converts the clause in place.
- **Category comparisons** become chips — click a value to swap it, and with `in (…)` chips toggle membership. The op dropdown offers `= ≠ in not in`.
- The **AND / OR pills** between rows are clickable — each one switches its own joiner in the text.
- **Brackets are real**: parenthesized groups render as indented blocks with their own *add condition* row (inserting inside the brackets), and the top-level *add* offers `and ( … )` to seed a new group or `( all ) or …` to wrap everything so far and start an alternative branch.

- Every row has a **× to remove it** — the clause and its joiner go together, removing the last condition inside brackets removes the whole group, and removing the only condition clears the filter.
- The drawer includes a **multiline editor** (with `#` comments) for the expression itself, live-validated; **reset** restores the filter as it was when the drawer opened.

Add a condition from the dropdown, then reshape it entirely with the widgets — nothing is locked in at add time.



The drawer projecting `FE > 40 and (LITO = "HEMATITE" or LITO = "CANGA")` — a slider with its operator dropdown, a bracketed group with clickable chips and an OR pill, a remove × on every row. Dragging the slider rewrites the expression above, live.

5 The ƒ workbench — derived columns

A derived column behaves exactly like a source column: the filter, colour ramps, stats, grade–tonnage, swath, the record panel, and every export see it. micro derives columns five ways, and each records *how* in the layer's provenance:

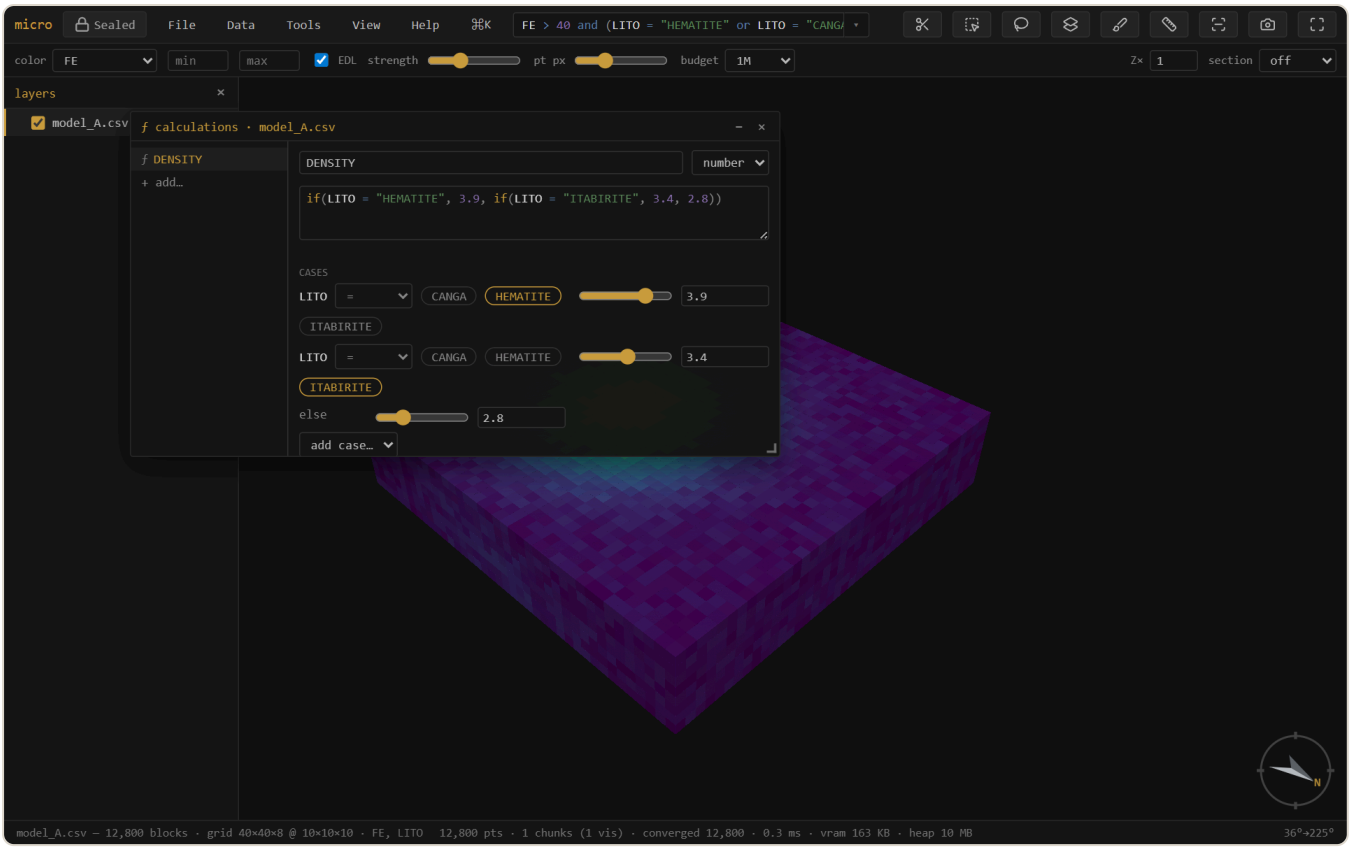
Kind	Made by	Stored as
ƒ calculated	an expression (`FE + 0.4 * MN`)	a definition — computed on the fly
materialized	freezing a ƒ, or an **interpolation** (NN / IDW with anisotropy) onto this layer	a Parquet column beside the source
painted ✎	brushing over the scene (`b`), or flag / assign-domains by a solid	a category column
joined	a key join or an own-lattice spatial join — see Join	materialized
broadcast	a collar column landed on drillhole intervals — see Drillholes	materialized

The Calculations window

Tools → Calculated columns… (or the **ƒ** button in Properties → columns) opens the calculations workbench: your ƒ columns listed on a rail, one editor on the right. The editor is multiline and syntax-highlighted, validates live against the layer's **full** schema — so a calculation can reference source columns, painted columns, materialized columns, **and other ƒ columns** (`FE_EQ` then `DBL = FE_EQ * 2`) — and the type is a visible choice (auto / number / text). Saving a definition immediately updates any live filter or colour that uses it.

Typing the expression is the primary interface, but below the editor the definition also **projects as widgets**, the same contract as the filter drawer:

- An `if(…)` ladder renders as a **case table** — condition | value, row per case, plus *else* and *add case*. Density by lithology (`if(LITO = "HEM", 3.9, if(LITO = "ITA", 3.4, 2.8))`) becomes a table you edit with chips and sliders.
- Any other expression lists its **numeric constants as sliders** — tune the coefficient in an equivalence formula and, if a filter or colour uses the column, watch the model respond live.



The ƒ workbench: a density-by-lithology `if()` ladder — the highlighted expression on top is the truth; below it the same definition as a case table of chips and value sliders.

Widget edits auto-save an existing column; typed edits wait for the explicit save. **materialize → stored** freezes a ƒ into a real Parquet column (it survives even if its inputs are later removed, and scans get faster); **save as recipe…** writes the whole column set as a re-runnable YAML — apply your standard derived columns to next month's model in one click.

Colour by ƒ

Pick `ƒ NAME` in the colour dropdown and micro computes the column once for display (a cached realization — recomputed automatically when the definition or anything it references changes, and never written to disk: the definition is the provenance). Combined with the constant sliders this closes a satisfying loop: drag the coefficient, the model recolours.

Interpolate onto a layer

Interpolate from this… (right-click the source points/holes) estimates a value onto **any** spatial target layer — nearest-neighbour or IDW, with a full anisotropy ellipsoid (ranges + dip-direction / dip / pitch), min/max samples, and expression filters on both the input samples and the output records. The result is a **materialized column on the target** — not a new layer — with the method and parameters recorded. Run several estimates as separate columns, then compose the final with a ƒ (`coalesce(AU_domA, AU_domB)`): every step stays auditable. If you

need a fresh lattice first, **New…** → **grid / block model** creates one, and interpolation targets it like any layer.

6 Drillholes

A drillhole set is **three tables bound by the hole id** — collars (one row per hole), survey (deviation stations), intervals (from/to samples) — and micro keeps that structure first-class rather than flattening it away.

The set in the layers panel

Every load lands as a **set group**: the group is the drillhole set, with a ◈ **collars** row (hole count), a ↧ **survey** row (desurvey method), and one interval layer per interval table. The rows are live — click one for its menu:

- ◈ **collars** — **Attribute table…** (the collar file as a table), **Filter holes…**, **Broadcast column to intervals**.
- ↧ **survey** — **Attribute table…**, **Re-desurvey…** (method: minimum curvature / balanced tangential / tangential; dip convention), **Show hole traces** (the full collar-to-EOH paths as thin grey sticks, filter-aware).
- The **set group** itself — hole traces, **Add interval table…** (assays + lithology + geotech share one geometry as sibling layers).

A drillhole set in the layers panel: the group is the set, with ◈ collars (hole count) and ↧ survey (desurvey method) rows above the interval layer. Click a row for its actions.

The hole filter — filter by collar data

Filter holes… takes an expression over the **collar** columns — `EOH > 200`, `COMPANY = "ACME"`, a campaign year — and applies the verdict through the whole set: only the matching holes' intervals stay visible (in every interval table at once), the traces redraw for the matching holes only, and the interval filter box then works *within* the kept holes. Nothing is copied or materialized — it's a live mask driven by the hole relation, and it persists with the project.

Broadcast — collar data on the intervals

Broadcast column to intervals lands a collar column on every interval, matched by hole id — so `EOH`, the logging company, or a campaign field becomes a real interval column you can

filter, colour, and export by. Numeric columns materialize to the project's column store; text columns become category columns with a legend. The operation is recorded with its inputs, like every derivation.

Display

Intervals render as true cylinders (**Stick radius…** sets the world-space radius). Colour by any interval column; sections cut the sticks true, and the set usually reads best with the model sectioned and the holes set to **don't cut**, threading whole through the opened deposit — which is exactly how the demo project arranges itself.

7 Selection & domaining

Select with `r` (rectangle) or `v` (lasso): what you *see* gets selected (occlusion-honest, all visible layers), tinted gold — **shift** adds, **alt** subtracts, `Esc` clears. The **select-through** toolbar toggle flips to the extruded tube: everything whose position falls inside the shape, occluded included.

Select by solid / surface (Tools menu, or right-click a mesh) classifies records against a closed mesh by winding number: **select** inside/outside, write a **flag column**, or a **fraction-inside** 0–1 column. Open surfaces (topo, weathering horizons) classify **above / below / between**. **Assign domains** runs the whole coded-model loop — pick the domain solids and a column name, and every block goes to the solid holding its largest fraction.

Paint a column (`b`) is interactive domaining by hand: a category column you fill by brushing over the scene. Painted values are real columns the filter, table, and export all see.

Selection → column… (right-click the layer, or the palette) materializes the current selection as an explicit **0/1 column** — every row gets a value, so it filters (`SEL = 1`), edits like any painted column, and rides every export. Selections themselves also persist in the project.

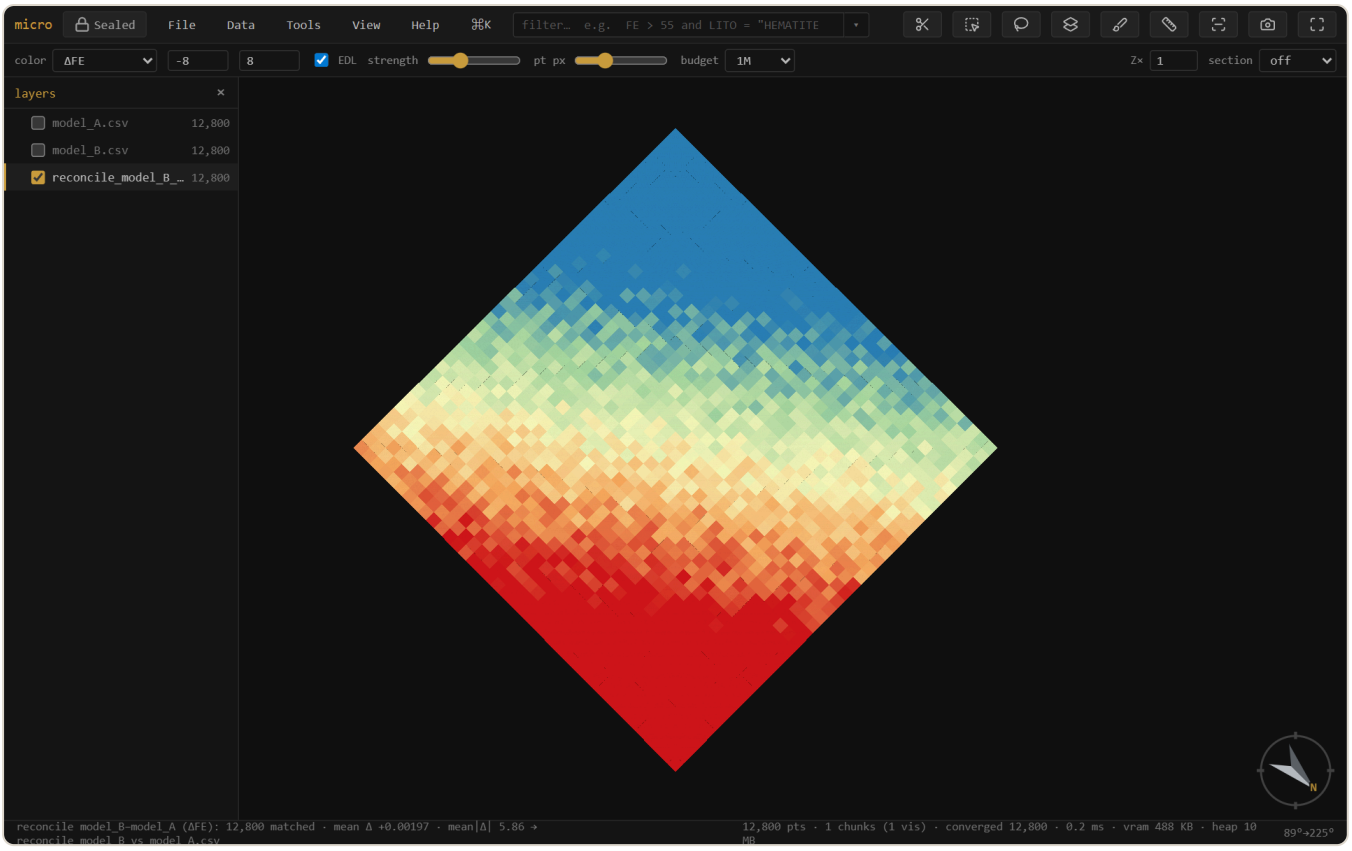
8 Join & reconcile

Two model versions, or a lookup table? The **join tab** (Properties → **join**) brings data across — and it follows one rule: **if the operation keeps this layer's records, the result lands as columns on it; only an operation that changes the record set makes a new layer.**

- **Key join → columns here** — match rows by a key column (a block id, a domain code) and the brought columns land on this layer: numeric as materialized columns, text as categories, name collisions prefixed. The classic domain→density lookup is one join, no new layer.
- **Spatial join → columns here** — with the target grid set to *this model's own lattice*, the other model's columns resample onto your existing cells (mean / sum / count, optionally weighted; categories by majority) and land as columns. A compatibility banner tells you whether the grids share a lattice, or why not.
- **→ new layer** — the secondary, export-shaped verb: a joined copy as a new layer/file. Spatial joins onto a *third* lattice (common-finest / coarsest) stay layers — they genuinely change the record set.

Reconcile is the preset: it produces a coloured **Δ map** (this model − reference) on a common lattice, with a diverging ramp centred on zero — blue where the new estimate is lower, red where it's higher. The summary reads **matched · mean Δ · mean|Δ| · added · removed**.

Sub-blocked models reconcile by **volume**: each variable-size block is aggregated onto the reference (parent) grid weighted by the space it fills, so a sub-blocked estimate compares cleanly against a regular one at the parent resolution.



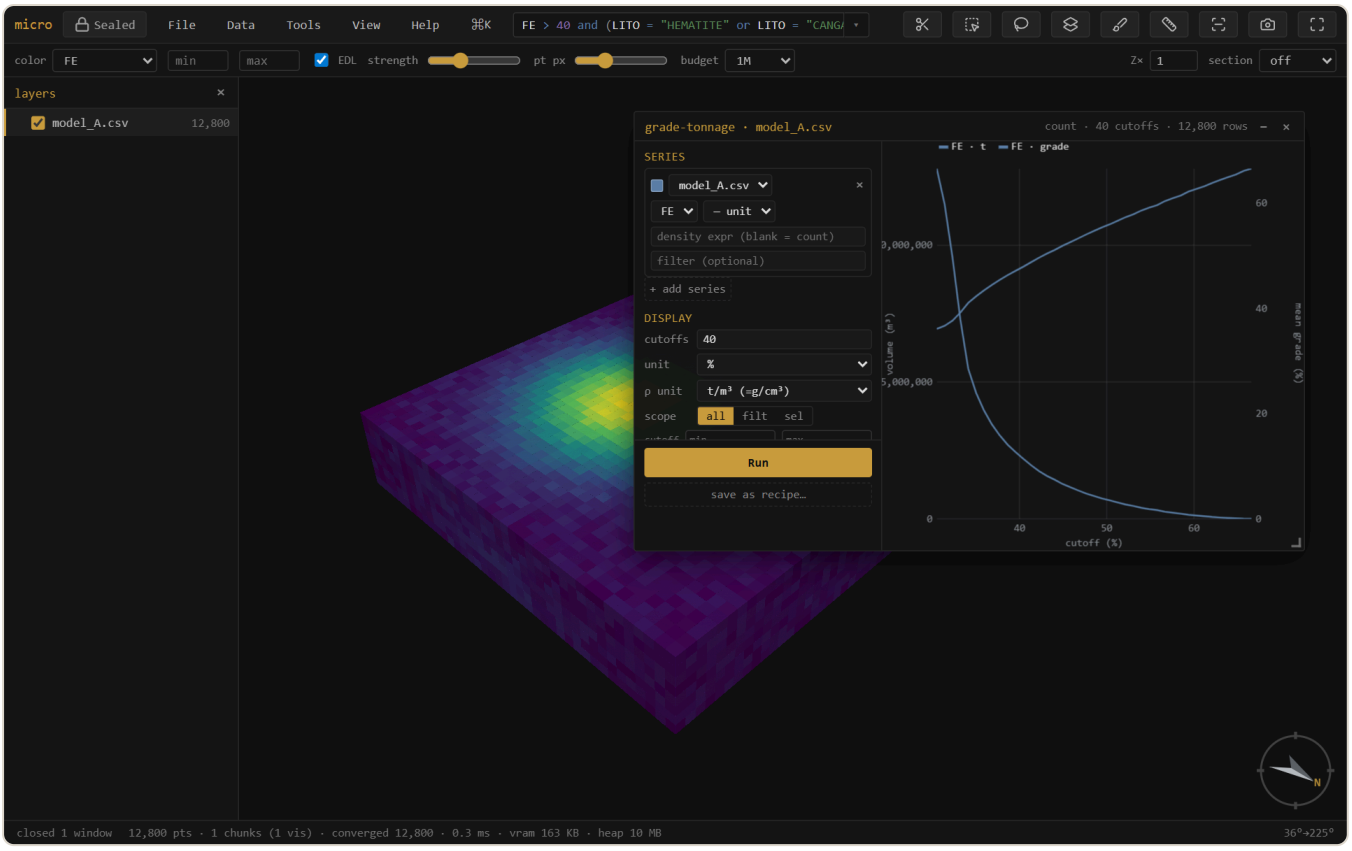
Reconciling two model versions. Global mean Δ is ≈ 0, but the Δ map exposes a clear **conditional bias** — the new estimate runs low in the north (blue) and high in the south (red). A global check would have missed it entirely.

> **Tip —** The Δ ramp auto-clips to the full range, which extreme blocks can widen. Tighten the **min/max** boxes (e.g. ±2× the mean|Δ|) to make the local pattern read, as above.

9 Grade–tonnage

Grade-tonnage… (right-click a model, or the palette) draws tonnage and mean grade above cutoff. It shares the swath's interface: a config rail of **series**, each one a **layer + grade column + declared unit + density expression + optional row filter** — configure, then **Run**. Because series pick their own layer, model-vs-model curves (this estimate against the last one) are one window; on a reconcile layer the reference/new pair is pre-seeded. Derived columns — calculated, materialized, joined — are all offerable as the grade or the density.

- **Density** is a free expression (`SG`, `2.7`, `SG * PARTIAL`) — tonnage = density × block volume, with a declared density unit converted to t/m³ so tonnes are tonnes. Blank = count/volume.
- **Units are declarations**, not display toggles: declare what each series' grade *is* (%, g/t, ppm, oz/t) and a **plot unit** — series convert onto the shared axis honestly, so a %-model overlays a ppm-sample set without lying.
- One streamed scan per Run caches each series' cumulative curve, so **cutoff count, plot unit, and manual axis ranges are live afterwards** — no re-scan.
- ↓ **png** saves the plot, ⧉ **copy** puts it on the clipboard, and ⧉ **table** copies the numbers as TSV — cutoff · tonnes · grade · contained metal per series, ready for a spreadsheet.



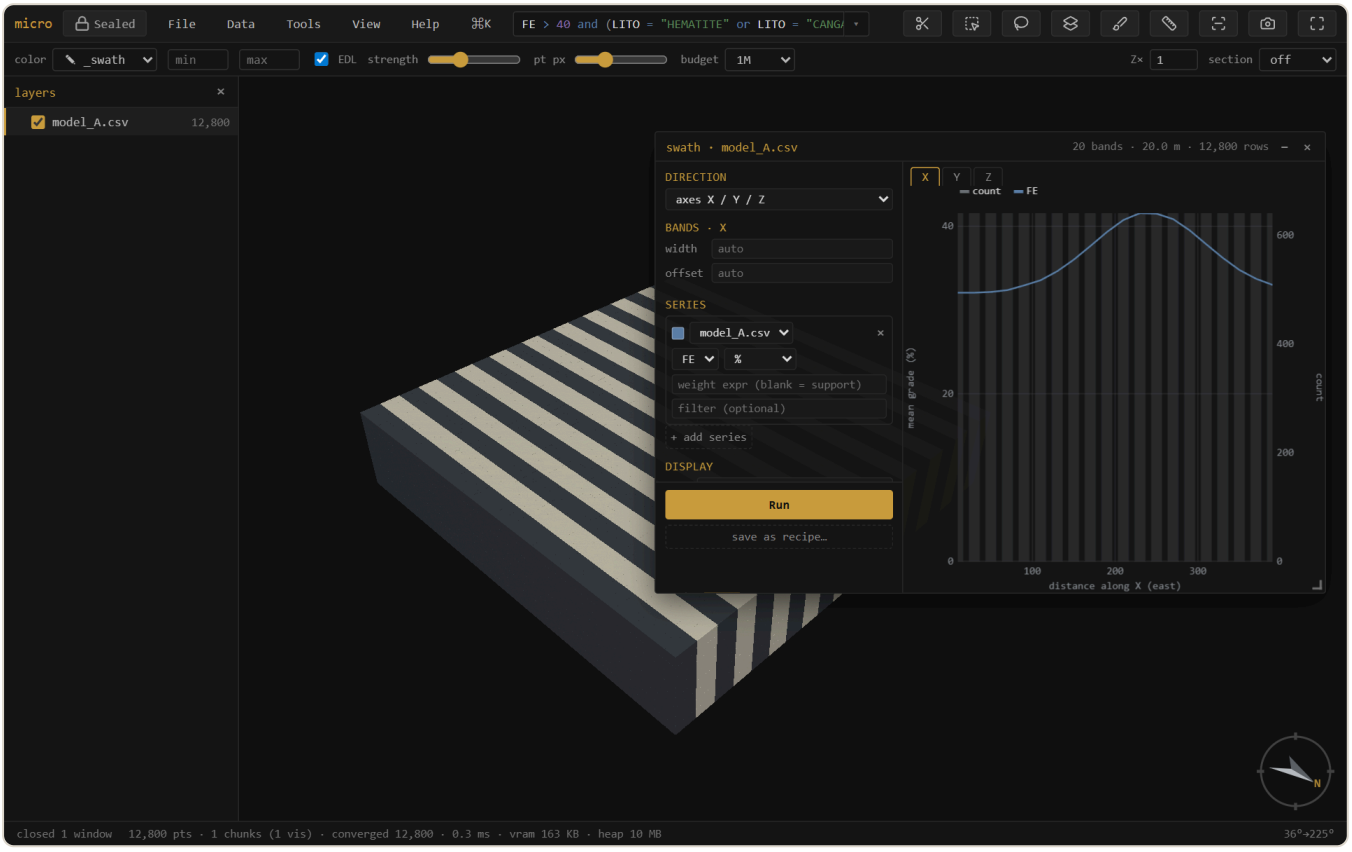
Grade–tonnage: tonnage (left axis) falls and mean grade (right axis) rises as the cutoff climbs — the recoverable-resource trade-off at a glance. The rail holds the series, units, density expression, scope, and manual ranges.

10 Swath / drift

Swath plot… shows mean grade per band along a direction — the classic drift check for local bias. The direction can be the grid **axes** X/Y/Z, an oriented trio from **dip-direction / dip / rake** (across-structure · along-strike · down-dip), or a single **trend / plunge**. Series work exactly like grade–tonnage's: layer + column + declared unit + a free **weight expression** (blank = block volume) + an optional per-series row filter — so kriging vs nearest-neighbour vs declustered samples overlay in one plot.

One **Run** streams everything once into fine bins; after that, **band width and offset re-bin instantly** — and they are **per direction**, so your E–W drift keeps its 25 m bands while the bench-by-bench Z profile keeps 10 m. Manual axis ranges, ⭳ png / ⧉ copy / ⧉ **table** (band · count · weight · per-series mean, TSV) match grade–tonnage.

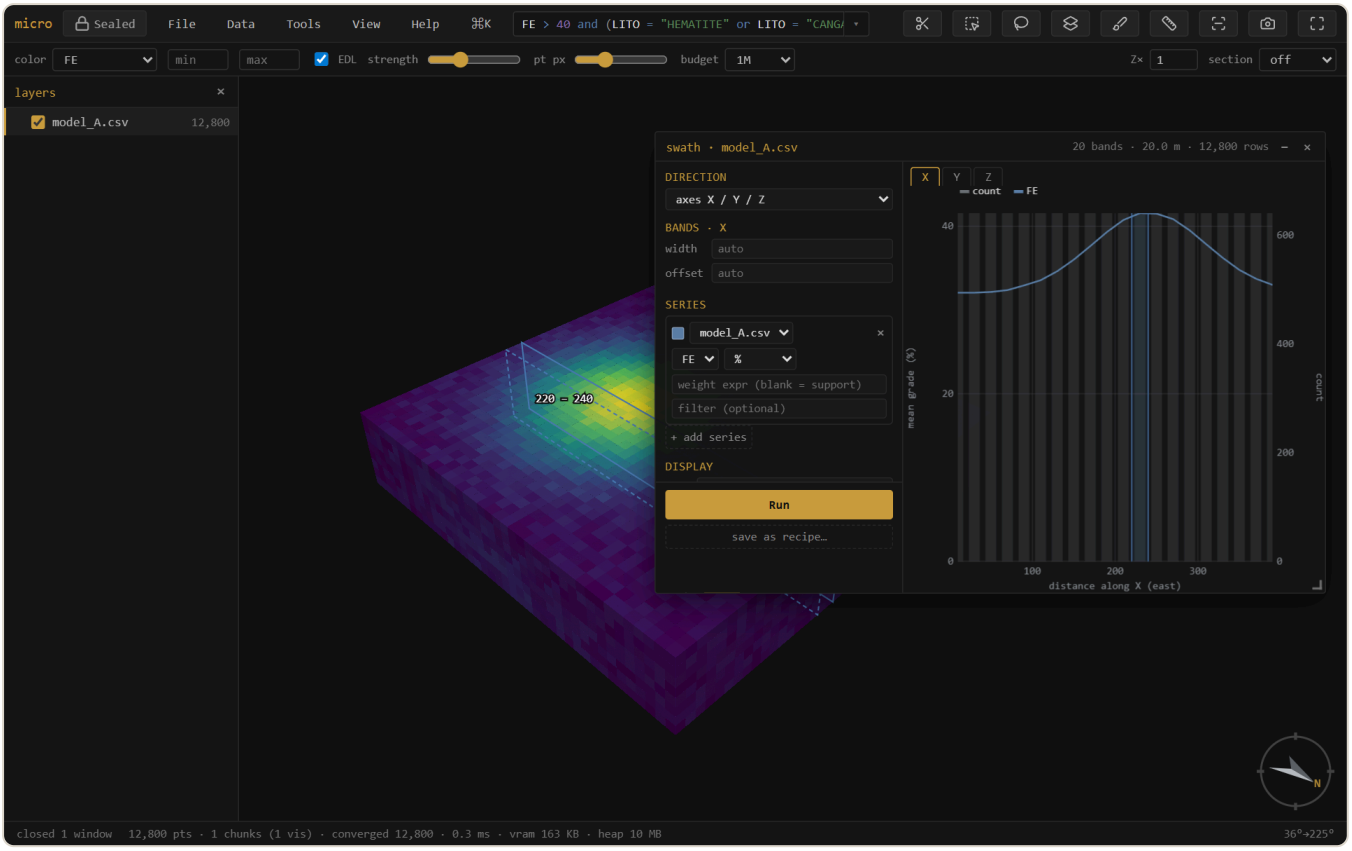
Zebra paints the alternate bands onto the 3D, so the slabs you see in space *are* the bins in the plot.



A swath along easting, with the bands zebra-striped onto the model. Because the plot and the stripes share one direction / width / offset, you read the profile against the deposit it came from.

The band ghost — hover the plot, see the slab

With **locate on 3D** on (it is by default), hovering the swath plot lights that band on the 3D as a translucent slab outline — so a dip in the profile points straight at the benches or the zone that caused it. **Drag** across the plot to hold a band *range* (violet), then turn it into a real 3D **selection** right from the menu that appears — and from there it's linked brushing, a 0/1 column, or an export like any other selection.

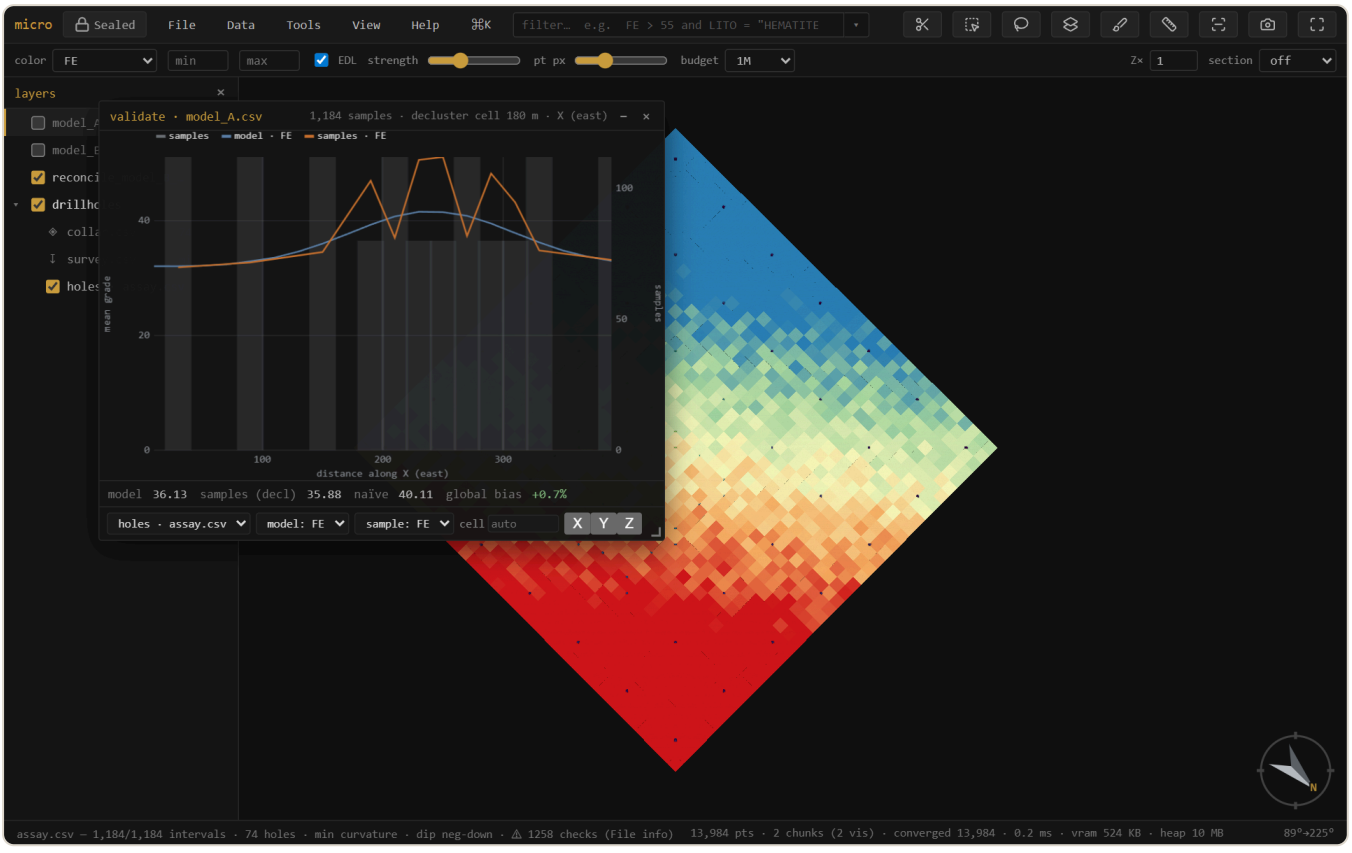


The band ghost: the bin under the cursor (shaded in the plot) is the slab outlined on the model. Drag a range and the menu offers *Select rows in…* — the plot becomes a selection tool.

11 Validate vs drillholes

Is the model consistent with the samples it was built from? **Validate vs drillholes…** takes a block model and a drillhole layer, **declusters** the sample grades (spatial sampling over-drills high grade, so a naïve sample mean is inflated), and reports the **global bias** (model mean vs declustered sample mean) plus a **model-vs-sample swath** along X/Y/Z.

> **Note —** The validation window is **temporarily hidden from the menus** while it's rebuilt on the new analysis chassis (per-series units, weight expressions, scopes). The machinery it uses — declustering, sample swaths — is unchanged; meanwhile the swath window already overlays model vs drillhole-sample series directly.



Validation. The naïve sample mean (40.1) is inflated by clustered infill drilling; **declustered** it's 35.9, essentially matching the model (36.1) — a **+0.7% global bias**. The swath compares model (blue) and sample (orange) grade along easting, slab by slab.

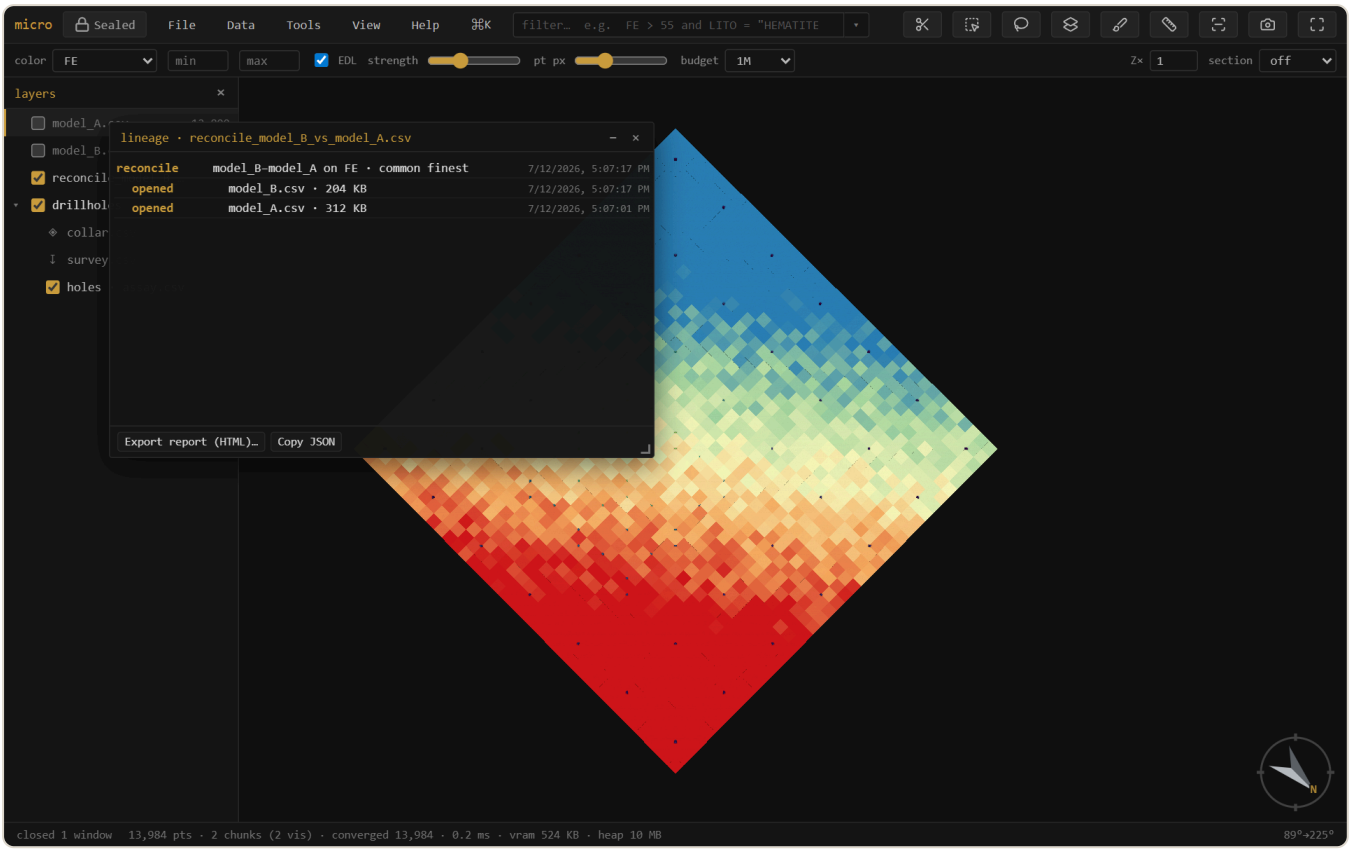
> **Note —** Samples are point support, blocks are block support, so a *global* grade offset between them can be real (change of support), not an error. The swaths are what flag *local*, conditional bias.

12 Linked brushing

The grade-tonnage and swath windows carry an **all / filt / sel** scope. **sel** follows the 3D selection: rubber-band a bench, a domain, or a suspect zone and the plot recomputes for just those blocks, live, as you brush. **filt** follows the **filter box** instead — type `LITO = "HEMATITE"` and every open analysis window scoped to it re-runs with each new expression. The analysis becomes interactive regional exploration, not a whole-model summary.

13 Lineage & provenance

Every derived layer *and every derived column* remembers *how it was made*. Opening a file is a leaf; a join, reconcile, or optimize wraps its source layers with the operation, its parameters, the build, and a timestamp — a tree. An estimated, broadcast, or joined column records its operation and inputs the same way, in the project's element manifests (see the project store). **Lineage…** shows the layer tree and exports a readable HTML **report** you can attach to a sign-off. Parquet exports carry the lineage embedded in the file, so the provenance travels *inside* the data.



The provenance tree for a reconciliation: the result wraps the two opened models. Export it as an HTML report, or read it back from any Parquet micro wrote.

14 Grids & surfaces

GeoTIFF / Surfer `.grd` / ESRI `.asc` open as heightfield layers — an elevation point cloud where cells are pickable records (huge DEMs decimate for display while the full grid stays for evaluation). A mesh right-click → **Rasterize to grid…** renders it onto a DEM (the surface↔grid converter, with fill / extrapolation for gaps). **New… → grid / block model** creates empty grids from parameters, covering a layer's bounds, or — **cover compatible models** — on a lattice that covers and is compatible with several models at once (the reconcile target). Grid layers export as ESRI ASCII.

Show a grid as a surface — relief, flat, and draped

Right-click a grid → **Reinterpret as** → **surface** renders it as a solid, smooth-shaded **relief** mesh — a DEM reads as terrain, not a gappy point cloud. In **Properties** → **Style** a surface then offers:

- **Drape** — colour it by its own *elevation/value*, or by **another loaded grid's values** (grade / geochem / slope over topo), sampled at each point. Paint the terrain with whatever channel you like. (The drape source is a 2D grid, sampled by position.)
- **Shape → flatten @ z** — turn a 2D data grid that has *no* elevation (a grade or geochem grid) into a **horizontal coloured sheet** at a chosen level, coloured by its value. A plan sheet, floating where you put it.
- **Band** — for a **multi-band GeoTIFF**, a picker to choose which band is shown (a GeoTIFF's bands are like a grid's columns). Switch it anytime; it re-reads that sample.

Surface, drape, flat, and band all **persist in the project** — reopen and the terrain comes back exactly as you left it. Sectioned surfaces draw their trace on the cut face (see Sections), and layer opacity applies — a translucent topo over an opened pit.

Surface tools: extend & combine

Extend to footprint… (right-click a surface — mesh or grid) grows it to cover a chosen extent: **nearest** holds the boundary z flat (the "extend the outer ring" move), **trend** continues the fitted dip. A topo that stops short of the model no longer leaves blocks unclassified — extend it, then flag above/below covers everything. **Combine surfaces…** rasterizes two or more surfaces onto one lattice and resolves the overlap by a rule: **min** (topo ∧ weathering base — take the lower), **max**, **first** (list order is priority — patch an end-of-month survey into the

original topo), or **mean**. Both produce **grid layers** — exact and fast for flags and reconcile — with the rule recorded in the layer name and its lineage; export .asc anytime.

15 The project store, inside

A micro project is a plain folder you can read without micro. This section is its anatomy — because auditability means being able to open the hood.

Parquet — the model store

Store as Parquet… (right-click a CSV/ `.dm` model) converts it Morton-sorted, so each row-group's footer becomes a tight spatial index and queries skip whole row-groups they can't match. The dialog says exactly what will happen — per-layer rows, sizes, and a **keep original** option — and the same conversion runs **when you save a project** (a one-time, remembered choice: always / ask / off). Painting stays editable through the reorder, the file self-describes (grid, extents, categories in its metadata, so it reopens with no discovery pass), and the whole thing round-trips. Parquet block models also open **streamed**, holding nothing decoded, with predicate and spatial push-down on the filter.

The element manifest — the data model on disk

Beside each data file the project writes `<file>.element.json`: a readable description of the element — its geometry interpretation, its record locations (a drillhole set lists *collars*, *vertices*, and one location per interval table, related by the hole id; a grid lists its lattice with coordinates implicit), its columns, and an **ops list**: every derivation (a calculation's expression, an estimate's method and parameters, a broadcast's source column, a join's inputs) recorded as data. The ops list is the audit trail — open the JSON and read exactly how every derived column came to be. It's also how the project restores: definitions reload from the manifest, and derived data that's missing or stale is simply re-derivable.

Column sidecars

Derived columns that carry data — estimates, materialized calculations, joins, broadcasts, painted domains — live in a `<file>.cols/` folder as **one Parquet file per column** (numeric as floats, categories as strings). Small, typed, self-describing, and readable by anything that reads Parquet.

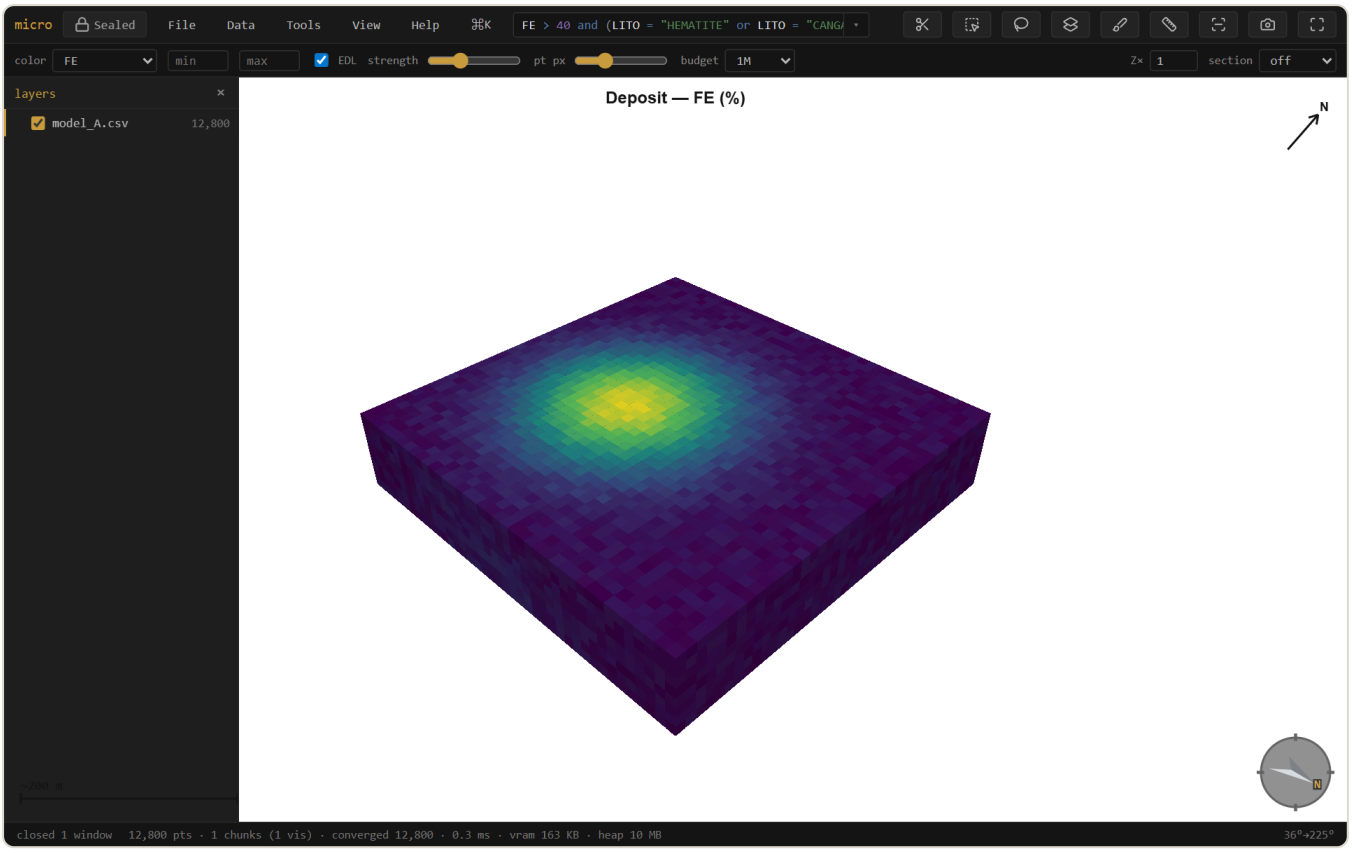
Sidecars & the memory budget

Files in a project also get a small `.meta.json` **sidecar** at save time: the discovery result (grid, extents, sub-blocking, categories — reopening skips the sweep entirely, even on a 13 GB `.dm`), column statistics, and a **band index** that lets filters skip stretches of CSV/`.dm` files that provably can't match. Sidecars are a cache: delete one and micro re-scans.

In memory, scanned columns **pin** into a budgeted cache (View → **Filter column cache**: off / auto / a size in MB — auto scales with your machine) so repeat filters and widget drags run with no file reads; derived-column values share the same budget and quietly reload from their sidecars when evicted. The memory panel (Help → memory) shows what's held.

16 Figures & decorations

View → Decorations & figure… toggles a **scale bar**, **north arrow**, colour **legend**, and **title** onto the viewport — they show live and are captured verbatim by **Export figure** (at 1× / 2× / 3× scale for print). Legends are **per layer** — stack one per model when several share the view, each with its own title. Set a **background** (white or paper for print; the decorations flip to dark ink to stay readable). The exported PNG is what you see, ready for a report — and it pairs with the lineage report for a fully auditable figure. The analysis plots have their own **↓ png / ⧉ copy / ⧉ table** buttons.



A report-ready figure: white background, scale bar, north arrow, grade legend, and title — composited onto the export exactly as shown.

17 Projects & export

In a Chromium browser, **File** → **Open project folder** keeps your layers, styling, filters, selections, camera, section, bookmarks & scenes — and your open **analysis windows** (see Windows) — in a plain folder (`project.json` + the data files). `Ctrl+S` saves.

The folder is organized **by data kind** — `models/` `drillholes/` `clouds/` `meshes/` `grids/` — with each file's manifest and sidecars beside it (see the project store). **Tidy project folder** migrates an older flat project in place, and **Scan folder for new data** treats the folder as an inbox: drop a file in by hand (or from another tool) and micro picks it up. A loose layer (opened from outside the folder) offers **Copy into project** from its context menu — always with a sized consent step, never silently.

Export rows… writes any layer back out — scope **all** / **filter matches** / **selection**, every column including calculated, painted, estimated, and joined ones, as CSV or Parquet. Opening a `.dm` and exporting *is* the `.dm` →CSV (or →Parquet) converter.

Bookmarks & scenes — save where you were looking

Two ways to save a view, under **View** → **Bookmarks** and **View** → **Scenes**. A **bookmark** is a *viewpoint* — camera + section — the quick nav aid: drop a few (**Save bookmark…**) and jump between them with the number keys `1`–`9`. A **scene** is the full *working state* — camera + section *plus* each layer's visibility, colour-by, and filter, so applying one puts the whole scene back the way it was (**Save scene…**, or **Save scene (with selection)…** to also pin the 3D selection). Applying a scene names any layer it wanted that isn't loaded, and skips it. Both live as small JSON files in the project folder — `bookmarks/` and `scenes/`, one per entry — so they travel with the project, are hand-editable, and get picked up by **Scan folder for new data**. There is no fly-through: applying a view snaps straight there.

Recipes — save an analysis to run again

save as recipe… (next to Run in the grade–tonnage, swath, and stats windows, in the Calculations window, and in the decorations panel) writes the current setup as a named YAML file in `recipes/` — every series, unit, weight expression, band, scope, and range; for calculations, the whole column set. A **figure recipe** captures the view + **section** + decorations + background + export scale; running it re-renders, exports the PNG, and restores everything it touched — the monthly plan-view figure regenerates itself. **Tools** → **Recipes** lists them; picking one opens the tool configured **and runs it**. Recipes are plain, commented, hand-editable files

— copy one between projects, keep an office standard set, drop one into the folder and **Scan folder** picks it up. Series reference layers **by name**; if the named layer isn't loaded, one question asks which loaded model to use instead — that's the whole parameter system. A recipe is data, never code: the strict YAML subset can't carry a script.

```
# micro recipe — grade-tonnage. Hand-edit freely; series reference layers by NAME
name: "Monthly GT"
tool: "gt"
params:
  nCut: 15
  gradeUnit: "pct"
  series:
    - layer: "model.parquet"
      col: "FE"
      weight: "SG"
```

Mesh export

Export mesh… (right-click a mesh layer; multi-selections and groups get **Export N meshes…**) writes **OBJ**, **PLY**, **MSH** (Leapfrog/ARANZ), or **LFM**. OBJ and LFM are multi-mesh — one named object per layer, with each layer's tint carried as the LFM colour — while PLY and MSH merge multiple layers into one mesh. Exports re-read the source geometry, so coordinates leave in **full double precision** (the PLY writer uses `double` — at UTM northings, float32 would cost half a metre). A Datamine wireframe pair in, an LFM out: micro is also a mesh-format converter.

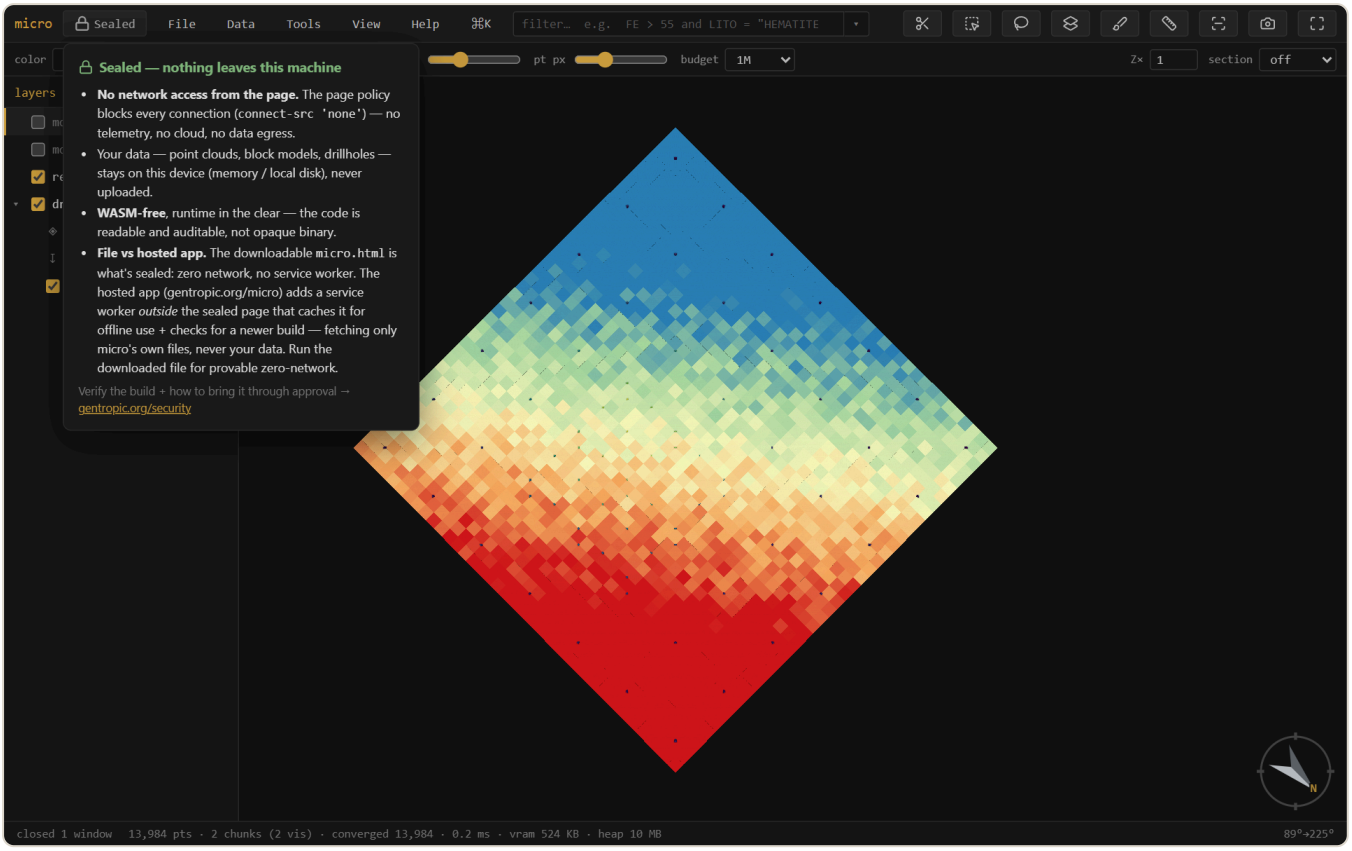
18 Windows

The analysis, table, calculations, and paint windows are all draggable and pinned to their layer (open several at once). **View** → **Windows…** lists every open window — click to focus, or **Cascade** and **Close all** to tidy up.

Grade–tonnage and swath setups **persist in the project**: every series, unit, weight expression, band width, scope, and manual range is snapshotted on each Run and on close, and a window left open when you save is **pinned** — reopening the project brings it back in place, configured, reading *restored — Run to compute*. Run stays manual, so a huge project never surprise-scans on open.

19 Security: Sealed

micro is **Sealed**: it makes **no network requests** — a browser security policy (`connect-src 'none'`) blocks every connection, so there is no telemetry, no cloud, and no auto-update. Your data — point clouds, block models, drillholes — stays on your device. The build is **WASM-free** with the runtime in the clear, so the code is readable and auditable rather than opaque binary.



The **Sealed** badge (top-left) explains the posture and links to the verifier. For confidential resource data, this is the point: nothing leaves the machine.

Verify the build and read how to bring it through approval at gentropic.org/security.

20 Install & offline

micro runs at gentropic.org/micro — install it as an app from the browser, or download `micro.html`: one file, no installer, that you can keep, e-mail, drop on a shared drive, or run fully offline by double-clicking. There is nothing to set up and nothing that phones home. Works in current Chrome / Edge / Firefox; project folders and mounts need a Chromium browser.

A Reference — keys & the expression language

Keyboard

Key	Action	Key	Action
`⌘/Ctrl K`	command palette	`/`	focus the filter
`f`	fit the data	`p`	layers panel
`n s e w`	look level that way	`d` / `u`	plan / from below
`o`	ortho / perspective	`x`	section on/off
`k`	knife — drag a section (⇧ snap 15°, ⇧⌥ 5°)	`l` / `⇧L`	face the section / its back
`,` `.`	step one section slab	`Ctrl wheel`	scrub the section
`Alt wheel`	slab thickness (thicker / thinner)	`1`–`9`	jump to bookmark
`m`	measure	`i`	file info
`r` / `v`	select rectangle / lasso	`b`	paint a column
`t`	attribute table	`[` `]`	step the active layer
`h` / `⇧H`	hide–show / solo the active layer	`F2`	rename the active layer
`⇧B`	block edges	`⇧P`	properties
`Ctrl S`	save project	`Ctrl Z`	undo paint
`Esc`	clear selection / cancel		

The expression language

One language everywhere — the filter, calculated columns, density and weight expressions, per-series filters, the hole filter. It is **total**: an expression can compare and compute but never loop, call out, or crash a scan — a blank input gives a blank result.

Comparisons	`> >= < <= = !=` · `between a and b` · `in (a, b, c)` · `contains "txt"` · `like "A%_1"` (SQL wildcards) · `matches "regex"` · `is blank` / `is filled`
Boolean	`and` · `or` · `not` — with parentheses for grouping
Arithmetic	`+ - * /` · `^` (power, Python-style precedence)
Numeric fns	`round int abs floor ceil mod log log10 exp sqrt pow min max clamp bin`
Blank handling	`coalesce(a, b, …)` (first filled) · `nullif(x, v)` (sentinel → blank: `nullif(AU, -99)`) · `ifnum isnum isblank isfilled` · the `blank` literal
Text fns	`upper lower trim len left right substr replace concat`
Conditional	`if(cond, then, else)` — nests into case ladders
Names	plain column names as-is (case-insensitive); anything with spaces or symbols in backticks: `` `Assay Au ppm` `` (the pandas convention)
Text values	in double quotes: `LITO = "HEMATITE"`
Comments	`# to end of line` — expressions may span multiple lines

`not` is a true complement: `not (AU > 5)` keeps blanks (they fail `AU > 5`), so it is *not* the same as `AU <= 5` — a deliberate, documented choice that makes filter + inverted-filter always cover the whole table. Autocomplete offers columns, functions, and category values — `Tab` / `Enter` accepts.

micro — part of the Auditable project, Geoscientific Chaos Union. Engine: `@gcu/condenser`; expressions: `@gcu/expr`; declustering: `@gcu/gsjs`. MIT.
Screenshots are generated from the app by an automated harness, so they track the current build. Docs for micro — see the in-app **Help** menu for the terse quick reference.